

**STATEMENT OF WORK (SOW) FOR
PREVENTATIVE MAINTENANCE, INSPECTION AND TESTING OF SPRINKLER
SYSTEMS, STANDPIPES SYSTEMS, BACK FLOW PREVENTERS, SPRINKLER
SYSTEM FIRE PUMPS AND FIRE HYDRANTS
AT USCG ACADEMY, NEW LONDON, CT**

1. INTRODUCTION

The United States Coast Guard Academy (CGA) is procuring services to conduct testing and inspections of the sprinkler, standpipe systems, fire hydrants, fire pumps and backflow prevention devices located in New London CT. All work shall be performed in accordance with the Statement of Work (SOW) and all current applicable State and Federal Regulations, National Fire Protection Agency Codes and USCGA Policies. For the purposes of this SOW, **NFPA current editions** are used. Changes to NFPA codes will require contractor to follow the current edition for the life of this contract.

2. BACKGROUND

2.1 The roots of today's Coast Guard were established in 1790 by Alexander Hamilton. Hamilton proposed the formation of the Revenue Marine, a seagoing military service that enforced customs and navigation laws, collected tariffs, hailed in-bound ships, made inspections and certified manifests. The first Coast Guard Academy (then called the Revenue Cutter School of Instruction) was held aboard the two-masted topsail schooner Dobbin. The first class of nine cadets boarded the Dobbin in Baltimore, Maryland in 1876 for a two-year training mission.

The Coast Guard Academy was a shipboard operation until 1890 when the first land-based campus was established in Curtis Bay, Maryland. In 1910, the Academy moved to the Revolutionary War fort and Army post at Fort Trumbull in New London, Connecticut. The modern Academy was born in 1915 with the merger of the Life Saving Service and Revenue Cutter Service. In 1932, the citizens of New London donated the present site to the Academy.

Today the Academy is committed to strengthening the nation's future by educating, training, and developing leaders of character who are ethically, intellectually, professionally, and physically prepared to serve their country and humanity, and who are strong in their resolve to build on the long military and maritime heritage and proud accomplishments of the United States Coast Guard.

The specifications contained herein are intended to describe the scope and nature of the work required of the Coast Guard Academy (CGA).

3. CONTRACTOR FURNISHED ITEMS

- 3.1 General. The contractor shall provide all equipment needed to service, test, repair, maintain, and inspect the systems covered by this contract.
- 3.2 Contractor shall be available to provide 24-hour emergency service to Coast Guard Academy in the event of activation, failure, or other circumstances with regards to sprinkler systems, fire hydrants and fire pumps on board CGA. Contractor shall respond within **four (4) hours** of service call for call deemed “**Emergency**” by CGA Fire Marshal. Emergency calls for service include broken sprinkler heads or piping, fire pumps inoperable or instances where there is a degradation of Life Safety systems that affect the safety of CGA personnel. Routine calls for service may be handled during the next available business day or at a mutually agreed time by the contractor and CGA Fire Marshal.
- 3.3 Contractor shall furnish all labor, materials, tools, equipment, supervision, and transportation necessary to perform Service Calls, Testing and Inspection as indicated in paragraph 3.2 and Section 4 of the Performance Work Statement. Funding for Service Call(s) and Repair Work will not be obligated upon award of the contract. Service Calls and Repair Work within the scope of this contract shall be conducted as a Task Order and funded on as needed basis.

4. INSPECTIONS

- 4.1 The contractor shall conduct a complete and comprehensive inspection of USCGA sprinkler systems, standpipe systems, and associated piping, valves and supervisory devices within **15 days of contract award**. A detailed report of existing conditions in each building will be submitted to the CGA Fire Marshal via hard copy and PDF on CD/DVD within **10 working days**, after completion of this inspection. Report will include make, model and date of manufacture for sprinkler heads, valves, and supervisory devices as well as any discrepancies found during inspections.
- 4.2 Four (4) quarterly inspections and testing of CGA sprinkler system and standpipe systems and all its components **IAW NFPA 25 (Current Edition)**. The results of the inspection and testing shall be documented in a detailed quarterly report. All Quarterly inspections **WILL** include a main drain test per CGA Fire Marshal. The main drain test may be waived by the CGA Fire Marshal if freezing temperatures exist and would create a hazardous situation to CGA personnel.
- 4.3 One (1) Annual inspection and testing of sprinkler and standpipe systems and their components **IAW NFPA 25 (Current Edition)**. The results of the additional testing and inspection will be included with the annual report.

NOTE: It is the responsibility of the contractor to ensure that an elevator contractor is available to provide access to elevator shafts for sprinkler system inspection and testing.

- 4.4 Monthly inspection of CGA backflow prevention devices **IAW NFPA 25 (Current Edition) 13.6.1**. See **Appendix B** for list of CGA backflow devices.
- 4.5 Annual backflow preventer inspection and testing **IAW NFPA 25 (Current Edition) 13.6.2**. The results of the inspection and testing shall be documented in an annual report. Backflow testing will include a forward flow test and a friction loss test on an annual basis. See **Appendix B** for list of CGA backflow devices.
- 4.6 Annual flow testing and inspection of ALL the CGA fire hydrants. This testing will include inspection and operation of Hydrant Control Valves along with recording hydrant static pressure and hydrant pitot pressure. See **Appendix C** for hydrant list.
- 4.7 Semi-annual maintenance of CGA Dry Sprinkler System compressors located in Chase Hall B Annex (2), Child Development Center (1), and Yeaton Hall (1). IAW manufactures recommendations and applicable standards.
- 4.8 Weekly running and testing of the Chase Hall A and Chase Hall E Annex fire pump IAW with manufacturer's recommendations and **IAW NFPA 25 (Current Edition) Chapter 8**.
- 4.9 Annual flow testing of the Chase A and Chase E Annex fire pump **IAW NFPA 25 (Current Edition) 8.3.3**.
- 4.10 All Inspection and Testing Reports shall be received at the below address within **10 calendar days** of inspection or testing completion. Inspection and testing reports shall include a written estimate to effect repairs for each discrepancy found.
- 4.11 All inspection and testing reports shall be in the following format: One (1) hard copy delivered via hand and one (1) PDF format delivered via email for all weekly, monthly, quarterly, semiannual and annual inspections.

Exception: The weekly fire pump reports shall be a PDF format delivered via email to the CGA Fire Marshal.
- 4.12 Weekly fire pump testing and inspection reports shall be received within **2 business days** of conducting the service. **ALL** discrepancies found shall be conveyed verbally to CGA Fire Marshal prior to contractor departing CGA.

- 4.13 Report routing. Route reports to:

USCG Academy (mee)
Attn: Fire Marshal
15 Mohegan Ave.
New London, CT 06320

5. ADDITIONAL ITEMS REQUIRED TO BE COMPLETED AT APPROPRIATE DATES DURING OPTION YEARS (See attachment I – USCGA Fire Sprinkler Schedule)

- 5.1 Conduct 5-year flow and hydrostatic test of ALL standpipe systems at the CGA, **IAW NFPA 25 (Current Edition) 6.3.1 and 6.3.2** and provide detailed report of conditions found.
- 5.2 Conduct 3-year trip test on CGA dry sprinkler systems located in Chase B (2), Child Development Center and Yeaton Hall. This test will serve as be performed **IAW NFPA 25 (Current Edition) 13.4.4.2.2.2** and every three years after.
- 5.3 Test anti-freeze solution in Chase Hall B Loading Dock **IAW NFPA 25 (Current Edition) 5.3.4**, adjust solution as necessary and provide report to CGA Fire Marshal. This is an annual requirement to be completed during annual inspection.
- 5.4 Conduct 5- year internal inspection of ALL piping, standpipe systems and FDC's at the CGA, **IAW NFPA 25 (Current Edition) 6.3.1 and 6.3.2** and provide detailed report of conditions found.

6. SCHEDULE OF INSPECTIONS

- 6.1 The first quarterly inspection will be conducted in the month of October.
- 6.2 The second quarterly inspection will be conducted in the month of January.
- 6.3 The annual inspection will be conducted in the month of April.
- 6.4 The third quarterly inspection will be conducted in the month of April.
- 6.5 The fourth quarterly inspection will be conducted in the month of July.
- 6.6 The backflow preventer test and inspection will be conducted in the month of July during the annual inspection.
- 6.7 Fire hydrant inspection, flush and flow testing will be conducted in the month of June.
- 6.8 Dry system compressor maintenance will be conducted in the months of February and August.
- 6.9 Annual flow test of the Chase Hall A and Chase Hall E fire pumps will be conducted in the month of July.
- 6.10 Running of the Chase Hall Echo Fire Pump will be conducted on a weekly basis with a minimum run time of **30 minutes**.
- 6.11 Running of the Chase Hall Alpha Fire Pump will be conducted on a weekly basis with a minimum run time of **10 minutes**.
- 6.12 Conduct monthly inspection of **ALL** CGA backflow prevention devices in conjunction with a weekly running of CGA fire pumps in Chase A and Chase E Annex.

6.13 Normal work hours are 7:30 a.m. to 4:00 p.m. Monday through Friday, excluding Federal Holidays and on days when special events are being held at the CGA. Inspections, maintenance, and repairs are to be performed during these hours. The contractor should coordinate the inspection schedule with the Coast Guard Academy's Fire Marshal.

7. BUILDINGS SPRINKLER SYSTEM INVENTORY: The Coast Guard Academy has the following systems: **twenty (20) wet, four (4) dry, twenty (20) standpipe systems and twenty (20) backflow prevention assemblies**. Each building is considered to have its own system. (The intent is to cover the PMS requirements for the entire Academy). Changes to current list of CGA systems due to construction or renovations will require a modification to the contract. See attached **Appendix A** for list of systems.

8. ADDITIONAL GENERAL REQUIREMENTS.

8.1 The contractor is to provide a copy of the company's current license (F1 Unlimited Contractor) and insurance and that of its employees conducting, inspection, testing and repairs under of the terms of this contract prior to the commencing of work under the contract.

8.2 Government Inspection. All work completed under this contract shall be subject to inspection by the Contracting Officer, or CGA Fire Marshal during and after completion of each task. Unsatisfactory work shall be corrected at no additional cost to the government.

8.3 Access to the Coast Guard Academy may vary depending on the existing state of alert. The contractor may expedite access to the Academy by providing a list of names to the Fire Marshal at least 24 hours prior to inspection. The Fire Marshal will forward the information to the Academy's security personnel.

8.4 Check-in and Check-out Procedure. At the beginning of each service visit to the Coast Guard Academy, Contractor personnel shall report in person at the CG Academy Fire marshal Office in the Sail Loft during normal work hours (7:30 am EST through 4:00pm EST). Contractor personnel will pick up necessary Chase Hall access card, keys, and a radio for communication. Contractor will provide a cell phone number to be used in the event radio communications are hampered by location. This radio/phone number will be used to maintain communications with CGA Police Central and CGA Fire Marshal, address questions or problems that may arise and update CGA Fire Marshal on work progress.

8.5 FAR Clause 52.222-41 requires the contractor to pay, at a minimum, the prevailing wage(s) and fringe benefits as determined by the Department of Labor's Wage Determination.

8.6 Contractor shall ensure that when water is discharged via hydrants, main drains or other means; **ALL** precautions are taken, and preventative measures are in place as to not disturb or destroy CGA grounds. Whenever possible, hoses and dispersing devices shall flow to large landscape areas (i.e., grass, shrubs, rocks, etc.) and not directly into

storm drains. Contractor **WILL** be responsible for making repairs and restoring CGA to its pre-testing/inspection condition.

8.7 This contract is a base plus four-year option.

Appendix A

1. **Hamilton Hall:** wet system, covering attic, **with one (1) DCVA** backflow preventer. The system has approximately 80 sprinkler heads. The sprinkler system and associated valves and switches are supervised by the fire alarm system.
2. **Chase Hall Alpha:** standpipe, wet system, covering attic, 1st, 2nd and 3rd floors of the North and South Wings, **with one (1) DC** backflow preventer and **electric** fire pump. There are approximately 700 sprinkler heads in the building and the entire system is supervised by the fire alarm system.
3. **Chase Hall Bravo:** standpipe, three (3) wet systems, and two (2) dry systems, covering all food service areas (including reefers and freezers), first floor, basement and sub basement areas with **one (1) Reduced pressure (RP)** backflow preventer. There are approximately 789 sprinkler heads covering the areas mentioned. The sprinkler system and associated valves are supervised by the fire alarm system.
4. **Chase Hall Charlie:** standpipe, wet system, covering the Uniform Shop, The system has approximately 50 sprinkler heads. The sprinkler system flow switch is monitored by the fire alarm system.
5. **Chase Hall Delta:** standpipe, wet system, covering attic, **with one (1) Double Check Valve Assembly (DCVA)** backflow prevention device. There are approximately 50 sprinkler heads covering the attic. The sprinkler system backflow device and flow switch are supervised by the fire alarm system.
6. **Chase Hall Echo:** standpipe wet system, covering entire building, **with two (2) DCVA** backflow preventers and **a diesel driven fire pump**. The building has approximately 460 sprinkler heads. Sprinkler system and fire pump are supervised by the fire alarm system.
7. **Chiller Plant:** wet system, covering entire building. There are approximately 17 sprinkler heads, and the system is supervised by the fire alarm system.
8. Eagle Pier: 5 YR flow and pressure test.
9. **Michel Hall (Medical):** standpipe, wet system, covering attic, **with one (1) DCVA** backflow prevention device. System has approximately 90 sprinkler heads, backflow prevention device and flow switch are supervised by fire alarm system.
10. **Leamy Hall:** standpipe, wet system, covering Auditorium, **with one (1) DCVA** backflow prevention device. There are approximately 25 sprinkler heads. The backflow device and flow switch are supervised by the fire alarm system.
11. **Johnson Hall:** standpipe, wet system, covering entire building (except garage area), **with one (1) RP** backflow preventer. The system has approximately 175 sprinkler heads. The system and associated valves are supervised by the fire alarm system.

12. **Ida Lewis Hall (Child Development Center):** wet and dry (1) systems covering the entire building and attic (dry), **with one (1) RP** backflow prevention device. System has approximately 390 sprinkler heads and supervisory devices are monitored by fire alarm system.
13. **Maintenance Shop:** standpipe, wet system, covering entire building, **with one (1) DCVA** backflow prevention device. The system has approximately 122 sprinkler heads. The sprinkler system and associated valves are supervised by the fire alarm system.
14. **Paint Shop:** wet system, covering entire building, **with one (1) DCVA** backflow prevention device. There are approximately 12 sprinkler heads, and the system is supervised by the fire alarm system.
15. **MCOE Hall:** standpipe, wet system, covering the first, second, third floor. workshop/paint booth areas, classrooms and attic. There is **one (1) DCVA** backflow prevention device. System has approximately 40 sprinkler heads. Backflow and flow switch are supervised by the fire alarm system.
16. **Yeaton Hall:** standpipe, wet system and one (1) dry system covering entire building, **with one (1) DCVA** backflow prevention device. System has approximately 575 sprinkler heads and, supervisory devices are monitored by fire alarm system. Dry system covers the attic while wet system covers the rest of the building. Backflow, flow switch and all associated valves are supervised by the fire alarm system.
17. **Roland Hall (Gym):** standpipe, wet system, covering entire building, **with one (1) DCVA** backflow prevention device. There are approximately 1650 sprinkler heads covering the entire building. All valves and switches are supervised by the fire alarm system.
18. **Munro Hall:** standpipe, (2) wet systems, covering entire building, **with one (1) RP** backflow prevention device. There are approximately 640 sprinkler heads. The backflow and associated valves are supervised by the fire alarm system.
19. **Satterlee Hall:** Standpipe, wet system, covering entire building, **with one (1) DCVA** back flow preventer. The system has approximately 550 sprinkler heads. The sprinkler system and associated valves and devices are supervised by the fire alarm system.
20. **Waesche Hall:** Standpipe, wet system **with one (1) DCVA** backflow prevention device. The backflow and flow switch are supervised by the fire alarm.
21. **Smith/Dimick Hall:** Standpipe **with one (1) DCVA** backflow prevention device. The backflow and flow switch are supervised by the fire alarm system.
22. **Alumni Center:** wet system **with one (1) DCVA** back flow preventer, covering entire building. The system has approximately 125 sprinkler heads. The system is supervised by the fire alarm system.

APPENDIX B

BACKFLOW PREVENTOR LIST

LOCATION	Size	Type	Model#	Serial#	
Alumni Center	6inch	Wilkins	350A	U02289	
Chase Bravo	6inch	Ames	4000SS	4CM0872	
Chase Charlie	4inch	Febco	825Y	12465	
Chase Delta	4inch	Ames/Colt	200A	LF-0393	
Chase Echo	8inch	Ames	200	HC-3091	
Chase Echo	.5inch	Watts	009 QT	18238	
Hamilton Hall	4inch	Ames	Colt 200DC	LC-1303	
Ida Lewis (CDC)	4inch	Watts	909	176740	
Johnson Hall	6inch	Febco	825YD	14787	
Leamy Hall	4inch	Ames	2000SS	169211- 0811	
Maintenance Shops	6inch	Ames	3000SS	3JJ1367	
Michel Hall	4inch	Ames/Colt	200A	LF-0403	
Munro Hall	6inch	Zurn/Wilkins	375	L00213	
Paint Shop	2.5inch	Ames/Colt	200A	LH-1913	
Pine Hall	4inch	Ames/Colt	200A	LH-0909	
Roland Hall	6inch	Ames	2000SS	2FM0146	
Satterlee Hall	4inch	Ames	2001SS	1001979	
Smith Hall	6inch	Ames/Colt	200DC	LE-0968	
Waesche Hall	4inch	Ames/Colt	200A	LH-0951	
Yeaton Hall	6inch	Ames	2000SS	2BM0364	
Chase A	8inch	Ames	200DC	KJ-0684	

Appendix C

CGA Fire Hydrant List

**Updated
06/30/17**

NUMBER (Color indicating flow rating)	Manufacturer	YEAR	LOCATION	Notes (*Date Installed)
H-1 (G)	Mueller		Leamy Hall, NW corner	*2012
H-2 (B)	Mueller		Smith Hall, SE corner	*2012
H-3 (G)	Mueller	2012	Munro Hall, by West entrance to 4th Deck	*2012
H-4 (O)	Mueller	2011	Chase "A" North on Bear Drive	* 2011
H-5 (G)	Mueller		Chase A, Henley Circle by Medical parking	* 2014
H-6 (O)	Mueller	2011	Satterlee Hall North on Bear Drive	* 2011
H-7 (G)	Mueller	2011	Tampa Dr. SE of Guard Shack	* 2011
H-8 (G)	Mueller	2011	Yeaton Hall South on Tampa Drive	* 2011
H-9 (G)	Mueller		East side of Satterlee Hall	*2012
H-10 (O)	Mueller		Seneca Drive between Quarters 1 & 2	* 2011
H-11 (O)	Mueller		Officer's Club main entrance	*2014
H-12 (O)	Mueller		Billard/Rolland, upper parking lot	*2014
H-13 (O)	Mueller	2011	Billard/Roland lower parking lot	* 2011
H-15 (G)	Mueller		Pine Hall, Southwest corner	*2012
H-16 (O)	Mueller	2014	McAllister Hall Southwest by parking lot	*2014
H-17 (G)	Mueller	2011	Public Works Shops, North	* 2011
H-18 (O)	Mueller	2011	Hamilton Hall North on Bear Drive	* 2011
H-19 (G)	Mueller	2011	Waesche Hall, SW hill	* 2011
H-20 (G)	Mueller	2011	Munro Hall, parking circle, South	* 2011
H-21 (G)	Mueller		Michel Hall, South	*2012
H-22 (B)	Mueller		McAllister Hall SE corner	*2014
H-23 (O)	Mueller		Harriet Lane and Bibb Street	*2012
H-24 (B)	Mueller	2011	Bibb St, South end of football field	* 2011
H-25 (O)	Mueller		Grounds Shop, South	* 2014
H-26 (G)	Mueller	2011	Baseball field, West side on Pickering	* 2011
H-27 (O)	Mueller		Visitor Center parking lot, North	* 2014
H-28 (O)	Mueller	2011	Seneca Drive between Quarters 3 & 4	* 2011
H-29 (G)	Mueller	1992	Chase A, Henley Circle by top of circle	
H-30 (G)	Mueller	2005	Rowing Center, SW	
H-31 (O)	Mueller	2005	Chase E, NE lower parking lot	
H-32 (G)	Mueller		Field House, South	* 2012
H-33 (O)	Mueller	2005	Chase E, NW, upper parking lot	
H-34 (B)	Mueller	2011	Alumni Center, road leading down	* 2011
H-36 (B)	Mueller	2012	Chiller Plant, Bibb & Pickering	